NATIONAL INSTITUTE OF TECHNOLOGY

Department of Computer Science and Engineering

Mid-Semester Examination, July-Dec 2026

B.Tech.	4th Semester	**DOE**	09th Oct 2026
Course Name	Data Structures	**Course Code**	CS031101/CS031201/CS031301
Maximum Time	2 hours	**Max. Marks**	30

Instruction:

1. Attempt all questions.
2. Assume any suitable data, if necessary.
3. The Marks, CO (Course Outcome) and BL (Bloom's Level) related to questions are mentioned on the right-hand side margin. [R: Remember, U: Understand, A: Apply, E: Evaluate, An: Analyze, C: Create]

Questions

#	Questions	CO	BL
1	**(i)** Apply the Quick Sort algorithm to sort the following list in alphabetical order: {D, A, T, A, S, T, R, U, C, T, U, R, E} Show all the steps involved in sorting the list. Also, draw the recursion tree representing the recursive calls made during the execution of the Quick Sort algorithm. **Assumption:** The last element of each subarray is selected as the pivot. **[4 M]**	CO_2, CO_4	A, E, An
	(ii) Write a C program to implement a Queue using Stacks. Also, determine the worst-case time complexity and perform an amortized analysis of the ENQUEUE(), DEQUEUE(), FRONT(), and ISEMPTY() operations. **[4M + 2M]**		
2	**(i)** Write a C program to find the middle element of a singly linked list without using any operation to count the total number of nodes. **[4M]**	CO_2, CO_4	R, U, A
	(ii) Consider the array: {64, 34, 25, 12, 22, 11, 90} Apply the Bubble Sort algorithm to sort the array in ascending order. Determine the contents of the array after the second iteration. Calculate the total number of comparisons and swaps performed till second iteration. **[3M]**		
	(iii) Write an algorithm to reverse the given singly linked list using a stack. **[3M]**		
3	**(i)** Explain Big (O), Omega (Ω), and Theta (Θ) asymptotic notations with suitable diagrams and examples. Also explain the concept of amortized analysis and illustrate it with a suitable example, clearly distinguishing it from worst-case analysis. **[5M]**	CO_1, CO_4	A, E
	(ii) Apply the master theorem to the following recurrence relations and find the asymptotic bounds: $T(n) = T(\sqrt{n}) + log(n)$ if $n \geq 2$, otherwise $O(1)$. **[3M]**		
	(iii) The keys 1, 3, 12, 4, 25, 6, 18, 20, 8 are inserted into an empty hash table of length 10 using open addressing with the hash function: $H(i) = i^2 \bmod 10$ and linear probing. Find: **[2M]**		
	a. The resultant hash table.		
	b. The maximum probe value.		